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SIMATS ENGINEERING

ASSESSMENT TOOL 1 – High (Coding Challenge)

COURSE CODE: CSA0904 COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO1: Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4).

QUESTIONS: 5 Total Marks: 100

ANNEXURE A Coding Challenge

Code of Conduct:

I G.MOHAMMAD SHAREEF(192311288)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

G.Mohammad shareef

Signature of the student:

1)Library Book Management using Classes & Encapsulatio:

CODE:

```
import java.util.*;

class Book {
    private int id;
    private String title;
    private boolean issued;

    Book(int id, String title) {
        this.id = id;
        this.title = title;
        issued = false;
    }

    int getId() {
        return id;
    }

    void issue() {
        if (!issued) {
            issued = true;
            System.out.println("Book Issued");
        } else
            System.out.println("Already Issued");
    }

    void ret() {
        issued = false;
        System.out.println("Book Returned");
    }

    void display() {
        System.out.println(id + " " + title + " " + (issued ? "Issued" : "Available"));
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        Book b[] = new Book[2];
        for (int i = 0; i < 2; i++) {
```

```

        System.out.print("Enter Book ID: ");

        int id = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Book Title: ");

        String title = sc.nextLine();

        b[i] = new Book(id, title);

    }

    b[0].issue();

    b[0].display();

    b[0].ret();

    b[0].display();

    System.out.print("Enter Book ID to Search: ");

    int s = sc.nextInt();

    for (Book x : b)

        if (x.getId() == s)

            x.display();

    }

}

```

Output:

The screenshot shows an IDE with a file named 'Main.java' and an 'Output' window. The code defines a 'Book' class with attributes 'id', 'title', and 'issued'. It includes methods for getting the ID, issuing a book, returning a book, and displaying book information. The main method interacts with the user to add books and search for them.

```

1- import java.util.*;
2- class Book {
3-     private int id;
4-     private String title;
5-     private boolean issued;
6-     Book(int id, String title) {
7-         this.id = id;
8-         this.title = title;
9-         issued = false;
10-    }
11-    int getId() {
12-        return id;
13-    }
14-    void issue() {
15-        if (!issued) {
16-            issued = true;
17-            System.out.println("Book Issued");
18-        } else {
19-            System.out.println("Already Issued");
20-        }
21-    }
22-    void ret() {
23-        issued = false;
24-        System.out.println("Book Returned");
25-    }
26-    void display() {
27-        System.out.println(id + " " + title + " " + (issued ? "Issued" : "Available"));
28-    }
29-}

```

The Output window displays the following text:

```

Enter Book ID: 101
Enter Book Title: plant
Enter Book ID: 102
Enter Book Title: animal
Book Issued
101 plant Issued
Book Returned
101 plant Available
Enter Book ID to Search:

```

2) 2. Employee Payroll System using Inheritance:

Code:

```
import java.util.Scanner;

class Employee {
    int id;

    String name;

    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }

    void calculateSalary() {
        System.out.println("Salary");
    }
}

class PermanentEmployee extends Employee {
    double basic;

    PermanentEmployee(int id, String name, double basic) {
        super(id, name);
        this.basic = basic;
    }

    void calculateSalary() {
        System.out.println("Permanent Employee Salary = " + (basic + basic * 0.20));
    }
}

class ContractEmployee extends Employee {
    int hours;
    double rate;

    ContractEmployee(int id, String name, int hours, double rate) {
        super(id, name);
        this.hours = hours;
        this.rate = rate;
    }

    void calculateSalary() {
```

```

        System.out.println("Contract Employee Salary = " + (hours * rate));
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.println("1.Permanent 2.Contract");
        int ch = sc.nextInt();
        Employee e;
        if (ch == 1) {
            System.out.print("Enter ID Name BasicSalary: ");
            e = new PermanentEmployee(sc.nextInt(), sc.next(), sc.nextDouble());
        } else {
            System.out.print("Enter ID Name Hours RatePerHour: ");
            e = new ContractEmployee(sc.nextInt(), sc.next(), sc.nextInt(), sc.nextDouble());
        }
        e.calculateSalary(); // Runtime Polymorphism
    }
}

```

OUTPUT:

The screenshot shows a Java IDE with a file named 'Main.java'. The code defines an abstract class 'Employee' with methods 'calculateSalary()' and 'getSalary()'. Two subclasses, 'PermanentEmployee' and 'ContractEmployee', override the 'calculateSalary()' method. The 'Main' class uses a 'Scanner' to take user input for the employee type (1 for Permanent, 2 for Contract), ID, name, and salary/basic pay, then creates an object and calls 'calculateSalary()'.

The output window shows the following execution results:

```

1.Permanent 2.Contract
1
Enter ID Name BasicSalary: 192324180
Balaji
200000
Permanent Employee Salary = 240000.0

```

At the bottom of the output window, it says: `*** Code Execution Successful ***`

3) 3. Vehicle Rental System using Runtime Polymorphism

Code:

```
import java.util.Scanner;

class Vehicle {
    void calculateRent(int days) {
        System.out.println("Rent");
    }
}

class Car extends Vehicle {
    void calculateRent(int days) {
        System.out.println("Car Rent = " + (days * 1000));
    }
}

class Bike extends Vehicle {
    void calculateRent(int days) {
        System.out.println("Bike Rent = " + (days * 500));
    }
}

class Bus extends Vehicle {
    void calculateRent(int days) {
        System.out.println("Bus Rent = " + (days * 2000));
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        Vehicle[] v = new Vehicle[3];
        v[0] = new Car();
        v[1] = new Bike();
        v[2] = new Bus();
        System.out.println("1.Car 2.Bike 3.Bus");
        System.out.print("Enter Choice: ");
        int ch = sc.nextInt();
        System.out.print("Enter Number of Days: ");
        int days = sc.nextInt();
```

```

        v[ch - 1].calculateRent(days); // Runtime Polymorphism
    }
}

```

Output:

```

Main.java
1- import java.util.Scanner;
2- class Vehicle {
3-     void calculateRent(int days) {
4-         System.out.println("Rent");
5-     }
6- }
7- class Car extends Vehicle {
8-     void calculateRent(int days) {
9-         System.out.println("Car Rent = " + (days * 1000));
10-    }
11- }
12- class Bike extends Vehicle {
13-     void calculateRent(int days) {
14-         System.out.println("Bike Rent = " + (days * 500));
15-     }
16- }
17- class Bus extends Vehicle {
18-     void calculateRent(int days) {
19-         System.out.println("Bus Rent = " + (days * 2000));
20-     }
21- }
22- public class Main {
23-     public static void main(String args[]) {
24-         Scanner sc = new Scanner(System.in);
25-         Vehicle[] v = new Vehicle[3];
26-         v[0] = new Car();
27-         ...
28-     }
29- }

```

Output

```

1.Car 2.Bike 3.Bus
Enter Choice: 1
Enter Number of Days: 3
Car Rent = 3000

--- Code Execution Successful ---

```

4) Bank Account System using Data Hiding

CODE:

```

import java.util.Scanner;

class BankAccount {
    private int accountNumber;
    private String holderName;
    private double balance;

    BankAccount(int acc, String name, double bal) {
        accountNumber = acc;
        holderName = name;
        balance = bal;
    }

    void deposit(double amt) {
        if (amt > 0)
            balance += amt;
    }

    void withdraw(double amt) {
        if (amt <= balance)
            balance -= amt;
        else

```

```

        System.out.println("Insufficient Balance");
    }

    void display() {

        System.out.println("Account No : " + accountNumber);

        System.out.println("Holder Name: " + holderName);

        System.out.println("Balance  : " + balance);}}

public class Main {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Account No: ");

        int acc = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Holder Name: ");

        String name = sc.nextLine();

        System.out.print("Enter Initial Balance: ");

        double bal = sc.nextDouble();

        BankAccount b = new BankAccount(acc, name, bal);

        System.out.print("Enter Deposit Amount: ");

        b.deposit(sc.nextDouble());

        System.out.print("Enter Withdraw Amount: ");

        b.withdraw(sc.nextDouble());

        System.out.println("\nAccount Details");

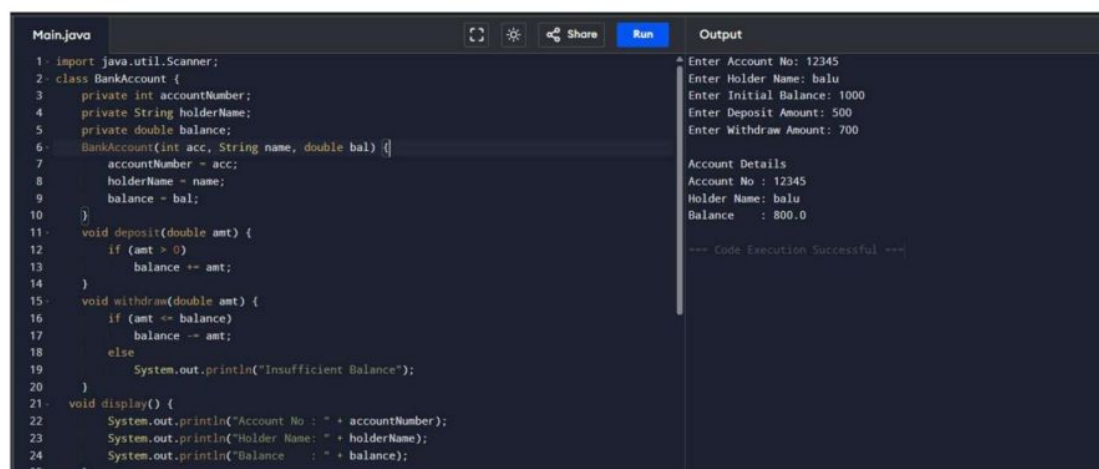
        b.display();

    }

}

```

OUTPUT:



The screenshot shows a Java IDE with a file named 'Main.java'. The code is as follows:

```

1. import java.util.Scanner;
2. class BankAccount {
3.     private int accountNumber;
4.     private String holderName;
5.     private double balance;
6.     BankAccount(int acc, String name, double bal) {
7.         accountNumber = acc;
8.         holderName = name;
9.         balance = bal;
10.    }
11.    void deposit(double amt) {
12.        if (amt > 0)
13.            balance += amt;
14.    }
15.    void withdraw(double amt) {
16.        if (amt <= balance)
17.            balance -= amt;
18.        else
19.            System.out.println("Insufficient Balance");
20.    }
21.    void display() {
22.        System.out.println("Account No : " + accountNumber);
23.        System.out.println("Holder Name: " + holderName);
24.        System.out.println("Balance  : " + balance);
25.    }
}

```

The output window on the right shows the following execution results:

```

Enter Account No: 12345
Enter Holder Name: balu
Enter Initial Balance: 1000
Enter Deposit Amount: 500
Enter Withdraw Amount: 700

Account Details
Account No : 12345
Holder Name: balu
Balance   : 800.0

--- Code Execution Successful ---

```

5) University Admission System using Method Overloading

CODE:

```
import java.util.Scanner;

class Admission {

    void calculateFee(double fee) {

        System.out.println("Undergraduate Fee = " + fee);

    }

    void calculateFee(double fee, int pg) {

        System.out.println("Postgraduate Fee = " + fee);

    }

    void calculateFee(double fee, double scholarship) {

        System.out.println("Scholarship Student Fee = " + (fee - scholarship));

    }

}

public class Main {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        Admission a = new Admission();

        System.out.println("1. Undergraduate");

        System.out.println("2. Postgraduate");

        System.out.println("3. Scholarship Student");

        System.out.print("Enter Choice: ");

        int ch = sc.nextInt();

        if (ch == 1) {

            System.out.print("Enter UG Fee: ");

            double fee = sc.nextDouble();

            a.calculateFee(fee);

        }

        else if (ch == 2) {

            System.out.print("Enter PG Fee: ");

            double fee = sc.nextDouble();

            a.calculateFee(fee, 1);

        }

        else if (ch == 3) {
```

```

        System.out.print("Enter Fee: ");

        double fee = sc.nextDouble();

        System.out.print("Enter Scholarship Amount: ");

        double s = sc.nextDouble();

        a.calculateFee(fee, s);

    }

    else {

        System.out.println("Invalid Choice");

    }

}

}

```

OUTPUT:

```

Main.java
1- import java.util.Scanner;
2- class Admission {
3-     void calculateFee(double fee) {
4-         System.out.println("Undergraduate Fee = " + fee);
5-     }
6-     void calculateFee(double fee, int pg) {
7-         System.out.println("Postgraduate Fee = " + fee);
8-     }
9-     void calculateFee(double fee, double scholarship) {
10-        System.out.println("Scholarship Student Fee = " + (fee - scholarship));
11-    }
12- }
13- public class Main {
14-     public static void main(String args[]) {
15-         Scanner sc = new Scanner(System.in);
16-         Admission a = new Admission();
17-         System.out.println("1. Undergraduate");
18-         System.out.println("2. Postgraduate");
19-         System.out.println("3. Scholarship Student");
20-         System.out.print("Enter Choice: ");
21-         int ch = sc.nextInt();
22-
23-         if (ch == 1) {
24-             System.out.print("Enter UG Fee: ");
25-             double fee = sc.nextDouble();
26-             a.calculateFee(fee);

```

```

Output
1. Undergraduate
2. Postgraduate
3. Scholarship Student
Enter Choice: 1
Enter UG Fee: 200000
Undergraduate Fee = 200000.0

=== Code Execution Successful ===

```

6) Shape Area Calculator using Abstract Class & Polymorphism

CODE:

```

import java.util.Scanner;

abstract class Shape {

    abstract void area();

}

class Circle extends Shape {

    double r;

    Circle(double r) { this.r = r; }

    void area() {

        System.out.println("Circle Area = " + (3.14 * r * r));

    }

}

```

```

class Rectangle extends Shape {
    double l, b;
    Rectangle(double l, double b) { this.l = l; this.b = b; }
    void area() {
        System.out.println("Rectangle Area = " + (l * b));
    }
}

class Triangle extends Shape {
    double base, height;
    Triangle(double base, double height) {
        this.base = base;
        this.height = height;
    }
    void area() {
        System.out.println("Triangle Area = " + (0.5 * base * height));
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        Shape s;

        System.out.println("1.Circle 2.Rectangle 3.Triangle");
        System.out.print("Enter Choice: ");
        int ch = sc.nextInt();

        if (ch == 1) {
            System.out.print("Enter Radius: ");
            s = new Circle(sc.nextDouble());
        } else if (ch == 2) {
            System.out.print("Enter Length and Breadth: ");
            s = new Rectangle(sc.nextDouble(), sc.nextDouble());
        } else {
            System.out.print("Enter Base and Height: ");
            s = new Triangle(sc.nextDouble(), sc.nextDouble());
        }
    }
}

```

```

        s.area(); // Polymorphism
    }
}

```

OUTPUT:

The screenshot shows a Java IDE with a file named 'Main.java'. The code in the editor is as follows:

```

26      System.out.println("Triangle Area = " + (0.5 * base * height));
27  }
28  }
29  public class Main {
30      public static void main(String args[]) {
31          Scanner sc = new Scanner(System.in);
32          Shape s;
33
34          System.out.println("1.Circle 2.Rectangle 3.Triangle");
35          System.out.print("Enter Choice: ");
36          int ch = sc.nextInt();
37
38          if (ch == 1) {
39              System.out.print("Enter Radius: ");
40              s = new Circle(sc.nextDouble());
41          } else if (ch == 2) {
42              System.out.print("Enter Length and Breadth: ");
43              s = new Rectangle(sc.nextDouble(), sc.nextDouble());
44          } else {
45              System.out.print("Enter Base and Height: ");
46              s = new Triangle(sc.nextDouble(), sc.nextDouble());
47          }
48
49          s.area(); // Polymorphism

```

The 'Output' pane on the right shows the following text:

```

1.Circle 2.Rectangle 3.Triangle
Enter Choice: 1
Enter Radius: 5
Circle Area = 78.5
--- Code Execution Successful ---

```

7. Hospital Management System using Interfaces:

CODE:

```

import java.util.Scanner;

interface MedicalRecord {

    void addRecord();

    void displayRecord();

}

class Patient implements MedicalRecord {

    int id;

    String name;

    public void addRecord() {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Patient ID: ");

        id = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Patient Name: ");

        name = sc.nextLine();

    }

    public void displayRecord() {

        System.out.println("Patient ID: " + id);

```

```

        System.out.println("Patient Name: " + name);
    }
}

class Doctor implements MedicalRecord {
    int id;
    String name;
    public void addRecord() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Doctor ID: ");
        id = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Doctor Name: ");
        name = sc.nextLine();
    }
    public void displayRecord() {
        System.out.println("Doctor ID: " + id);
        System.out.println("Doctor Name: " + name);
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        MedicalRecord m;

        System.out.println("1. Patient");
        System.out.println("2. Doctor");
        System.out.print("Enter Choice: ");
        int ch = sc.nextInt();

        if (ch == 1)
            m = new Patient();
        else
            m = new Doctor();

        m.addRecord();
    }
}

```

```

        m.displayRecord();
    }
}

```

OUTPUT:

```

Main.java
1 import java.util.Scanner;
2 interface MedicalRecord {
3     void addRecord();
4     void displayRecord();
5 }
6 class Patient implements MedicalRecord {
7     int id;
8     String name;
9     public void addRecord() {
10         Scanner sc = new Scanner(System.in);
11         System.out.print("Enter Patient ID: ");
12         id = sc.nextInt();
13         sc.nextLine();
14         System.out.print("Enter Patient Name: ");
15         name = sc.nextLine();
16     }
17     public void displayRecord() {
18         System.out.println("Patient ID: " + id);
19         System.out.println("Patient Name: " + name);
20     }
21 }
22 class Doctor implements MedicalRecord {
23     int id;
24     String name;
25     public void addRecord() {
26         Scanner sc = new Scanner(System.in);

```

```

Output
1. Patient
2. Doctor
Enter Choice: 1
Enter Patient ID: 123
Enter Patient Name: balu
Patient ID: 123
Patient Name: balu

--- Code Execution Successful ---

```

8) Online Shopping Order System using Packages

CODE:

```

import java.util.Scanner;
class Product {
    String name;
    double price;
    Product(String name, double price) {
        this.name = name;
        this.price = price;
    }
}

class Order {
    Product p;
    int qty;
    Order(Product p, int qty) {
        this.p = p;
        this.qty = qty;
    }
    void generateBill() {
        double total = p.price * qty;
        System.out.println("\n----- BILL----- ");
        System.out.println("Product : " + p.name);
    }
}

```

```

        System.out.println("Price : " + p.price);
        System.out.println("Quantity : " + qty);
        System.out.println("Total : " + total);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Product Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Product Price: ");
        double price = sc.nextDouble();
        System.out.print("Enter Quantity: ");
        int qty = sc.nextInt();
        Product p = new Product(name, price);
        Order o = new Order(p, qty);
        o.generateBill();
    }
}

```

OUTPUT:

```

Main.java
1- import java.util.Scanner; class Product {
2-     String name;
3-     double price;
4-     Product(String name, double price) {
5-         this.name = name;
6-         this.price = price;
7-     }
8- }
9- class Order {
10-     Product p;
11-     int qty;
12-     Order(Product p, int qty) {
13-         this.p = p;
14-         this.qty = qty;
15-     }
16-     void generateBill() {
17-         double total = p.price * qty;
18-         System.out.println("\n----- BILL -----");
19-         System.out.println("Product : " + p.name);
20-         System.out.println("Price : " + p.price);
21-         System.out.println("Quantity : " + qty);
22-         System.out.println("Total : " + total);
23-     }
24- }
25- public class Main {

```

```

^ Enter Product Name: Laptop
Enter Product Price: 300000
Enter Quantity: 2

----- BILL -----
Product : Laptop
Price : 300000.0
Quantity : 2
Total : 600000.0

=== Code Execution Successful ===

```

9) Student Result Analyzer using Arrays of Objects

Code:

```
import java.util.Scanner;

class Student {
    int roll, total;
    String name;
    int m1, m2, m3;
    double avg;

    Student(int roll, String name, int m1, int m2, int m3) {
        this.roll = roll;
        this.name = name;
        this.m1 = m1;
        this.m2 = m2;
        this.m3 = m3;
    }

    void result() {
        total = m1 + m2 + m3;
        avg = total / 3.0;

        System.out.println("\nRoll No : " + roll);
        System.out.println("Name : " + name);
        System.out.println("Total : " + total);
        System.out.println("Average : " + avg);

        if (avg >= 90)
            System.out.println("Grade : A");
        else if (avg >= 75)
            System.out.println("Grade : B");
        else if (avg >= 50)
            System.out.println("Grade : C");
        else
            System.out.println("Grade : Fail");
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
```

```

System.out.print("Enter Number of Students: ");

int n = sc.nextInt();

Student s[] = new Student[n];

int max = 0;

for (int i = 0; i < n; i++) {

    System.out.println("\nEnter Roll No, Name, Marks1, Marks2, Marks3:");

    s[i] = new Student(sc.nextInt(), sc.next(), sc.nextInt(), sc.nextInt(), sc.nextInt());

    s[i].result();

    if (s[i].total > s[max].total)

        max = i;

}

System.out.println("\nClass Topper: " + s[max].name);

System.out.println("Top Marks: " + s[max].total);

}

}

```

Output:

```

Main.java
1- import java.util.Scanner;
2- class Student {
3-     int roll, total;
4-     String name;
5-     int m1, m2, m3;
6-     double avg;
7-     Student(int roll, String name, int m1, int m2, int m3) {
8-         this.roll = roll;
9-         this.name = name;
10-        this.m1 = m1;
11-        this.m2 = m2;
12-        this.m3 = m3;
13-    }
14-    void result() {
15-        total = m1 + m2 + m3;
16-        avg = total / 3.0;
17-        System.out.println("\nRoll No : " + roll);
18-        System.out.println("Name : " + name);
19-        System.out.println("Total : " + total);
20-        System.out.println("Average : " + avg);
21-        if (avg >= 90)
22-            System.out.println("Grade : A");
23-        else if (avg >= 75)
24-            System.out.println("Grade : B");
25-        else if (avg >= 50)
26-            System.out.println("Grade : C");

```

```

Output
Enter Number of Students: 3

Enter Roll No, Name, Marks1, Marks2, Marks3:
123
balu
60
70
85

Roll No : 123
Name : balu
Total : 215
Average : 71.66666666666667
Grade : C

Enter Roll No, Name, Marks1, Marks2, Marks3:
124
yashu
80
50
62

Roll No : 124
Name : yashu
Total : 192
Average : 64.0

```

10) Smart Home Device Controller using Hierarchical Inheritance:

CODE:

```
import java.util.Scanner;

class Device {
    void on() {
        System.out.println("Device ON");
    }
    void off() {
        System.out.println("Device OFF");
    }
    void power() {
        System.out.println("Power Consumption");
    }
}

class Light extends Device {
    void power() {
        System.out.println("Light Power = 20 Watts");
    }
}

class Fan extends Device {
    void power() {
        System.out.println("Fan Power = 75 Watts");
    }
}

class AirConditioner extends Device {
    void power() {
        System.out.println("Air Conditioner Power = 1500 Watts");
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        Device d;
```

```

        System.out.println("1.Light");
        System.out.println("2.Fan");
        System.out.println("3.Air Conditioner");
        System.out.print("Enter Choice: ");
        int ch = sc.nextInt();

        if (ch == 1)
            d = new Light();
        else if (ch == 2)
            d = new Fan();
        else
            d = new AirConditioner();

        d.on();
        d.power(); // Runtime Polymorphism
        d.off();
    }
}

```

OUTPUT:

The screenshot shows a Java IDE with a file named 'Main.java'. The code defines a base class 'Device' with methods 'on()', 'off()', and 'power()'. It also defines three subclasses: 'Light', 'Fan', and 'AirConditioner', each overriding the 'power()' method to print their respective power consumption (20 Watts, 75 Watts, and 1500 Watts respectively). The main method in 'Main.java' prompts the user to enter a choice (1 for Light, 2 for Fan, 3 for Air Conditioner). In the output window, the user has entered '2', and the program has successfully executed, displaying the output for the Fan class.

```

Main.java
1- import java.util.Scanner;
2- class Device {
3-     void on() {
4-         System.out.println("Device ON");
5-     }
6-     void off() {
7-         System.out.println("Device OFF");
8-     }
9-     void power() {
10-        System.out.println("Power Consumption");
11-    }
12- }
13- class Light extends Device {
14-     void power() {
15-         System.out.println("Light Power = 20 Watts");
16-     }
17- }
18- class Fan extends Device {
19-     void power() {
20-         System.out.println("Fan Power = 75 Watts");
21-     }
22- }
23-
24- class AirConditioner extends Device {
25-     void power() {
26-         System.out.println("Air Conditioner Power = 1500 Watts");
27-     }
28- }
29-
30- public class Main {
31-     public static void main(String[] args) {
32-         Scanner sc = new Scanner(System.in);
33-         Device d = null;
34-         System.out.println("1.Light");
35-         System.out.println("2.Fan");
36-         System.out.println("3.Air Conditioner");
37-         System.out.print("Enter Choice: ");
38-         int ch = sc.nextInt();
39-
40-         if (ch == 1)
41-             d = new Light();
42-         else if (ch == 2)
43-             d = new Fan();
44-         else
45-             d = new AirConditioner();
46-
47-         d.on();
48-         d.power(); // Runtime Polymorphism
49-         d.off();
50-     }
51- }

```

Output

```

1.Light
2.Fan
3.Air Conditioner
Enter Choice: 2
Device ON
Fan Power = 75 Watts
Device OFF

--- Code Execution Successful ---

```